Project Report

Paving the Road to Wisdom

A Weighted Network Analysis of the English Wikipedia Philosophy Sink

Submitted by

Group 6

Sneha Nagmoti 12342090

Snehal Suhane 12342100

Bonta Aalaya 12340530

Kaki Venkata Vaneesha 12240740

Indian Institute of Technology Bhilai

Subject Network Science

Contents

1. Introduction

This project investigates the *"All Roads Lead to Philosophy"* phenomenon—the widely observed behaviour wherein clicking the first non-parenthetical hyperlink in any Wikipedia article, and repeating the process on each subsequent article, eventually leads to the *Philosophy* page. While commonly treated as a trivia curiosity, this study approaches it as a rigorous network-science problem by modelling the English Wikipedia as a **directed, scale-free graph**.

The distinctive contribution of this work lies in overlaying real-world **human-attention metrics** (monthly Pageview counts) onto the structural skeleton of the hyperlink graph. This augmented representation allows us to distinguish between two fundamentally different drivers of knowledge convergence:

- **Preferential Attachment** — popularity-driven navigation, in which the "first-link" structure of Wikipedia merely mirrors collective reading habits;
- **Functional Necessity** — structural bottlenecks that every convergent path must traverse regardless of the traffic volume they attract.

By quantifying both forces, we aim to determine whether the Philosophy sink is a reflection of how humans naturally organise and consume knowledge, or merely an artefact of Wikipedia's editorial conventions.

2. Research Questions & Hypotheses (Tentative)

The study is organised around twelve high-level theories drawn from Network Science. Each question is paired with an operationalised measurement approach and an interpretive framework.

2.1 The "Main Street" vs. "Back Alley" Hypothesis

Question. Do the paths to Philosophy rely predominantly on high-traffic hubs, or does the network's structure compel users to traverse obscure, low-traffic articles?

Approach. We compare **Betweenness Centrality** with **Pageview Weight** for every node on sampled Philosophy-converging paths. Nodes that exhibit high centrality but low pageview counts are identified as **"Silent Pillars"**—articles that are structurally indispensable to the All-Roads phenomenon yet are rarely read by actual Wikipedia users.

Interpretation. If high-centrality nodes are also high-traffic, the phenomenon is a feature of human navigation. If Silent Pillars are common, the phenomenon is a structural compulsion that operates largely below the threshold of ordinary readership.

2.2 The "Rich-Get-Richer" (Preferential Attachment) Hypothesis

Question. Is the network's convergence structure ultimately a popularity contest?

Approach. We calculate the Pearson correlation coefficient r between a node's **In-Degree** (the number of distinct articles that list it as their first valid link) and its **Pageview Weight**.

Interpretation. A value of $r \approx 1$ suggests that Wikipedia is a popularity-driven system in which the "first link" rule merely encodes collective reading preferences. A low r reveals the existence of Silent Pillars—structurally vital but largely unread articles—demonstrating that convergence operates independently of human attention.

2.3 The Pageview Gradient (Uphill vs. Downhill)

Question. Is the journey to Philosophy a "downhill" slide from niche, low-traffic articles toward broad, highly popular concepts, or an "uphill" climb against the grain of human attention?

Approach. We analyse the **Pageview Gradient** ΔW along every sequential edge in every sampled path. Specifically, for each path $P = (v_1, v_2, \ldots, v_n)$ where v_n is *Philosophy*, we compute ΔW between v_i and v_{i+1}.

Interpretation. If ΔW is consistently positive, the structure of Wikipedia functions as an **attention funnel** toward General Knowledge. A random gradient implies that the link structure is largely independent of what people actually choose to read.

2.4 Network Robustness: The "Achilles' Heel"

Question. If we surgically remove the top 1% of articles ranked by pageview weight (a **Targeted Attack**), does the All-Roads-to-Philosophy phenomenon survive?

Approach. We iteratively remove the highest-pageview nodes, re-run path-following experiments on the pruned graph, and measure changes in network **Diameter** and sink-convergence rate after each removal round.

Interpretation. If convergence collapses quickly, the phenomenon is a fragile construct held together by a small number of **"Super-Roads"**—articles such as *Science* or *Language*.

If it survives, convergence is a robust, emergent property of the entire encyclopedia.

2.5 Community Detection: "Cultural" vs. "Academic" Gatekeeper Hubs

Question. Do different thematic neighbourhoods of Wikipedia (Science, Pop Culture, History) use different Gatekeeper articles to reach Philosophy?

Approach. We apply the **Louvain** or **Leiden** community-detection algorithm to the weighted graph, identifying clusters by link-density and pageview flow. We then examine which articles serve as inter-community bridges.

Interpretation. If communities are topically coherent (e.g., Pop Culture always passes through *Social Science*; Physics always passes through *Mathematics*), then the Gatekeeper structure reflects human-curated subject taxonomy. If bridges are topically arbitrary, convergence is a purely structural property.

2.6 The Semantic Boundary Test

Question. Is convergence to Philosophy driven by semantic similarity between topics, or purely by the structural rule of following the first link?

Approach. Let X denote the partition of nodes obtained via community detection on the first-link graph, and let Y denote the partition induced by Wikipedia categories. We quantify their alignment using Normalised Mutual Information (NMI):

$$\mathrm{NMI}(X, Y) = \frac{2I(X;Y)}{H(X) + H(Y)}.$$

Interpretation. High NMI indicates that detected communities closely match semantic categories, suggesting that paths remain within topical boundaries for most of their trajectory. Low NMI implies weak alignment, meaning that convergence to Philosophy cuts across categories early, reflecting a primarily structural (rather than semantic) process.

2.7 Degree Correlations (Assortativity)

Question. Does popularity seek popularity?

Approach. We compute the **Assortativity Coefficient** r based on pageview-weighted degree sequences to determine whether high-traffic pages predominantly link to other high-traffic pages (*assortative*) or to obscure bridge articles (*disassortative*).

Interpretation. In many social networks, popular nodes cluster together. If the Wikipedia philosophy graph is **disassortative**, it means high-traffic pages systematically point to obscure bridge pages to get to Philosophy—a qualitatively distinct navigational behaviour from social-network patterns.

Interpretation. If older pages are more central, foundational knowledge articles drive convergence. If current popularity (pageview weight) is the better predictor, human reading behaviour dominates structural importance.

2.8 Philosophical Imposters

Question. Which nodes reach Philosophy in ≤ 5 steps despite originating from semantically distant categories?

Approach. Let $G = (V, E)$ be the directed graph induced by the Wikipedia first-link rule, and let $f : V \to \mathcal{C}$ map articles to categories. For each node $v \in V$, define its hitting time to Philosophy $T(v)$. We identify the set

$$I = \{v \in V \mid T(v) \leq 5 \ \wedge \ d_{\mathcal{C}}(f(v), f(\text{Philosophy})) \gg 0\},$$

where $d_{\mathcal{C}}$ measures inter-category distance. We further detect paths exhibiting large discrete jumps in category space:

$$\exists\, k \text{ such that } d_{\mathcal{C}}(f(v_k), f(v_{k+1})) \text{ is large.}$$

Interpretation. Nodes in I are "imposters": their small $T(v)$ arises from abrupt category transitions rather than gradual abstraction, reflecting structural artefacts in first-link selection.

2.9 The Barabási–Albert Model: Preferential Attachment & Page Age

Question. Is structural importance on the path to Philosophy determined by a node's *historical age* or its *current popularity*?

Approach. The Barabási–Albert (BA) model predicts a "First-Mover Advantage": older nodes accumulate disproportionate connectivity over time. We test this by correlating each article's **creation date** (available in the Wikipedia dump) with its structural distance to Philosophy and its betweenness centrality on the path graph.

Interpretation. If older pages are more central, foundational knowledge articles drive convergence. If current popularity (pageview weight) is the better predictor, human reading behaviour dominates structural importance.

3. Tech Stack & Infrastructure (Tentative)

Building and analysing a graph of this scale (4.7 million+ nodes, tens of millions of edges) requires a specialised, memory-efficient stack to avoid prohibitive RAM usage on consumer hardware.

3.1 Language & Runtime

- `Python 3.12+`

3.2 Data Engineering

- `xml.etree.ElementTree` (`iterparse`): Event-driven streaming parser for massive Wikipedia XML dumps. Processes records sequentially without loading the entire file into memory.
- `mwparserfromhell`: High-speed parser for complex Wikitext markup, used to resolve templates, strip infoboxes, and identify the first valid internal link.
- `bz2` (standard library): Reads compressed Pageview archives in `.bz2` format directly in binary-stream mode without prior extraction, preserving disk space.

3.3 Graph Engine & Analysis

- `NetworkX`: Primary graph library for computing centrality measures (betweenness, in-degree), network diameter, motif detection, community detection, and Markov-chain simulation.
- `Pandas`: Statistical correlation, dataframe operations, and Pageview data filtering and joining.
- `SciPy`: Pearson/Spearman correlation tests, entropy calculations, and null-model statistical comparisons.

3.4 Visualisation

- `Gephi`: Rendering the final macroscopic layout of the Philosophy Sink, producing high-resolution force-directed maps of the weighted graph (ForceAtlas2 layout algorithm).

- `Matplotlib/Seaborn`: Plotting Pageview Gradients, Degree Distributions, correlation scatter plots, and entropy curves.

3.5 Intermediate Storage & Checkpointing

- `Pickle`: Serialising intermediate Python objects (dictionaries, graph snapshots) during pipeline development.
- `Parquet` (`PyArrow`): Column-oriented storage for large dataframes (node-attribute tables), enabling fast random-access queries without re-parsing source dumps.
- `GEXF`: Final graph serialisation format (XML-based), preserving both topology and node attributes for consumption by both NetworkX and Gephi.

3.6 Development Environment

- IDE: Visual Studio Code (Windows 11 environment)

4. Methodology & Data Pipeline

4.1 Architectural Overview

Dealing with a dataset of this magnitude—where the English Wikipedia XML dump alone is over **100 GB uncompressed**—requires a fundamental shift from *loading* data to *streaming* data. Standard methods such as `pandas.read_csv()` would exhaust available RAM and crash the system within seconds of execution.

The project therefore adopts a **Linear-Pass Streaming Architecture**: the entire encyclopaedia is never held in memory; the pipeline processes exactly one article page at a time, discards it immediately, and moves to the next. This architecture guarantees a peak memory footprint below 500 MB regardless of input file size.

4.2 Stage 1 — The "Straw" Method (Compressed Streaming)

Rather than unzipping the 20 GB+ `.bz2` dump files to disk before reading, we use Python's built-in `bz2` and `gzip` libraries to open and read the files in **binary-stream mode**. Conceptually, this resembles drinking a thick milkshake through a straw: only one small chunk of bytes is in transit at any moment. The CPU decompresses data in real-time, keeping peak RAM usage consistently below 500 MB regardless of the source file's uncompressed size.

4.3 Stage 2 — The Structural Skeleton Pass (XML Parsing)

The decompressed byte stream is fed into `xml.etree.ElementTree.iterparse()`. Unlike conventional XML parsers that construct a full in-memory document tree, `iterparse` is **event-driven**: it fires a callback event each time it encounters a closing `</page>` tag, processes that single article's XML node, and immediately discards it via `element.clear()`. This critical step wipes the article from RAM before the next one arrives.

Within each page event, the following preprocessing steps are applied in order before any link is extracted:

1. **Redirect Resolution.** All redirect articles are resolved to their canonical targets before any link is recorded, maintaining node integrity across the graph.
2. **Template & Caption Stripping.** All Wikitext templates (including infoboxes) and Image/File captions are stripped via `mwparserfromhell` logic, preventing spurious first-link assignments.
3. **Parenthesis Erasure.** A depth-counter algorithm erases all text within parentheses (including nested parentheses), faithfully implementing the *"first non-parenthetical link"* rule.
4. **First Valid Internal Link Identification.** After all preprocessing, the parser identifies the first valid `[[Internal Link]]` and records it as the directed edge (Source $\rightarrow$ Target) for that article node.

4.4 Stage 3 — Popularity Injection (Pageview Mapping)

The Wikimedia Pageviews archive (`.bz2` format) is a separate and substantial dataset. It is streamed independently and filtered specifically for `en.wikipedia` entries. The result is stored as a **Key-Value Map** (Python dictionary) of the form:

$$\{ \text{"Article Title"} \mapsto \text{View Count} \}$$

Because titles are strings and counts are integers, a dictionary of 6 million entries fits into roughly 2–4 GB of RAM—manageable on most modern laptops. This popularity map is subsequently used to inject Pageview counts as node attributes into the graph.

4.5 Stage 4 — Graph Synthesis (NetworkX)

Once the full link list and the popularity map are available, a `networkx.DiGraph()` object is initialised. Every (Source, Target) pair is added as a directed edge, and

`.set_node_attributes()` is called to inject the Pageview counts from the popularity map onto the corresponding nodes.

The resulting weighted directed graph $G = (V, E)$ is defined as:

- V: the set of all Wikipedia article nodes, each annotated with a pageview weight $w(v)$;
- E: the set of directed first-link connections $\{(u, v) \mid v \text{ is the first valid link in article } u\}$.

The graph is serialised to `.gexf` format (preserving topology and attributes) and checkpointed in Parquet/Pickle so downstream analyses can be re-run without reprocessing the 100 GB source file.

4.6 Tools for Large-Scale Execution

Table 1 summarises the tool choices made specifically to support the scale and constraints of this pipeline.

Table 1: Tool selection rationale for the large-scale execution pipeline.

Task	**Tool / Library**	**Rationale**
lightblue Parsing	`LXML` / `ElementTree`	Fast, low-level C-based XML parsing. Critical for processing the 100 GB dump without performance bottlenecks on consumer hardware.
Wikitext Logic	`mwparserfromhell`	Handles the "messy" Wikitext edge cases—templates, file links, infoboxes—that plain regular expressions cannot reliably resolve.
lightblue Intermediate Storage	`Pickle` or `Parquet`	Saves the constructed graph as a checkpoint so individual analyses can be re-run without re-parsing the full 100 GB source dump each time.
Graph Analysis	`NetworkX`	Industry-standard Python library for centrality metrics, diameter calculation, community detection, and motif analysis.
lightblue Visualisation	`Gephi`	Renders large-scale force-directed graph layouts with node-weight and community colouring for publication-quality output.

4.7 Analytical Procedures

The following analyses are performed on the fully constructed and attribute-annotated graph G:

1. **Global Metrics.** In-Degree distribution, network diameter, average path length to

Philosophy, and degree distribution (tested against a power-law fit).

2. **Centrality Analysis.** Betweenness Centrality for all nodes on sampled Philosophy-converging paths; Spearman correlation with Pageview weights to identify Silent Pillars.

3. **Preferential Attachment Test.** Pearson correlation r between In-Degree and Pageview weight; bootstrapped confidence intervals.

4. **Pageview Gradient Analysis.** ΔW computed along every edge of every sampled path; mean gradient trajectory plotted against hop distance from Philosophy.

5. **Robustness / Targeted Attack.** Iterative removal of top-1% pageview nodes; re-measurement of diameter and convergence rate after each removal step.

6. **Community Detection.** Louvain / Leiden algorithm on the weighted graph; NMI comparison between detected communities and Wikipedia Categories.

.

7. **Assortativity.** Assortativity Coefficient computed from pageview-weighted degree sequences using the Newman formula.

8. **Page-Age Correlation.** Regression of article creation date against structural distance to Philosophy and betweenness centrality; controlling for pageview weight.

9. **Philosophical Imposters.** Path-length comparison against Erdős–Rényi and configuration-model baselines; identification of statistical outliers by Z-score.

5. Expected Contributions & Outcomes

Upon completion, this project is expected to deliver the following concrete artefacts and findings:

1. A fully annotated weighted directed graph (`.gexf`) of the English Wikipedia philosophy-sink subgraph, comprising approximately 6 million nodes with associated pageview weights—a resource that can be reused by the network-science community.

2. A quantitative determination of whether the All-Roads-to-Philosophy phenomenon is primarily driven by **Preferential Attachment** (human reading behaviour) or

Functional Necessity (structural bottlenecks), based on the Pearson correlation between In-Degree and Pageview weight.

3. A catalogue of **Silent Pillars**—articles with high betweenness centrality but low pageview weight—that are structurally indispensable to the convergence path yet practically invisible to ordinary readers.

4. A **Robustness Report** quantifying how the philosophy-convergence property degrades (or survives) under targeted removal of high-traffic nodes, with implications for the resilience of knowledge organisation systems.

5. A **Semantic Boundary Analysis** (NMI score) quantifying how much the convergence path respects versus ignores Wikipedia's subject-category boundaries, revealing whether the phenomenon is disciplinarily structured or purely topological.

   .

6. A high-resolution **Gephi visualisation** of the Philosophy Sink, colour-coded by community membership and sized by pageview weight, suitable for inclusion in a research publication.

6. Limitations & Scope Boundaries

- **Language Scope.** This study analyses the English Wikipedia exclusively. Cross-lingual generalisability is not claimed; the structural properties of the Philosophy sink may differ substantially in other Wikipedia editions.
- **Temporal Snapshot.** The graph is constructed from a single dump snapshot. The longitudinal stability of convergence paths is explored as a secondary question (see Section **??**) but is not the primary focus.
- **Out-Degree Constraint.** The first-link rule constrains every node's out-degree to exactly 1 (excluding self-loops and "red links"). Certain standard network metrics (e.g., standard betweenness centrality) must therefore be adapted for this constrained topology.
- **Pageview Proxy.** Pageview counts are used as a proxy for human attention but are subject to confounding from bot traffic, automated link-following scripts, and seasonal spikes (e.g., news events driving sudden article popularity).

- **First-Link Ambiguity.** For a small fraction of articles, the definition of "first non-parenthetical link" is ambiguous due to unusual Wikitext formatting. Our parser resolves these conservatively, but edge-case handling introduces a small measurement uncertainty.
- **Memory vs. Scale Trade-off.** The streaming architecture imposes a one-pass constraint on the raw dump. Multi-pass algorithms (e.g., exact betweenness centrality on the full graph) are approximated using sampling strategies, introducing bounded error.

References

[1] Barabási, A.-L. (2016). *Network Science*. Cambridge University Press. http://networksciencebook.com/

[2] Wikimedia Foundation. (2024). *English Wikipedia Database Dumps*. https://dumps.wikimedia.org/

[3] Wikimedia Foundation. (2024). *Wikimedia Pageviews Archive*. https://dumps.wikimedia.org/other/pageviews/

[4] Fortunato, S. (2010). Community detection in graphs. *Physics Reports*, 486(3–5), 75–174. https://doi.org/10.1016/j.physrep.2009.11.002

[5] Milo, R., Shen-Orr, S., Itzkovitz, S., Kashtan, N., Chklovskii, D., & Alon, U. (2002). Network Motifs: Simple Building Blocks of Complex Networks. *Science*, 298(5594), 824–827. https://doi.org/10.1126/science.298.5594.824

[6] Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Wiley-Interscience.

[7] Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

[8] Newman, M. E. J. (2002). Assortative Mixing in Networks. *Physical Review Letters*, 89(20), 208701. https://doi.org/10.1103/PhysRevLett.89.208701

[9] Earwig. (2024). *mwparserfromhell: A Python parser for MediaWiki wikicode* (v0.6.6). https://github.com/earwig/mwparserfromhell